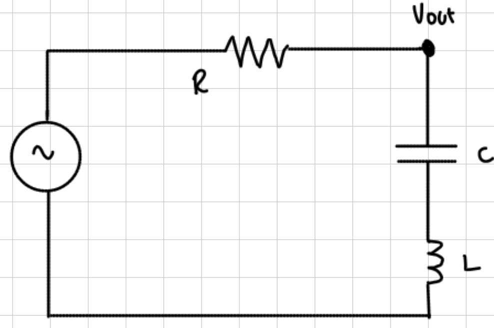


$R = 100\,K\ \Omega$
$C = 3.18\ \mu F$
$L = 31{,}83\ mH$

$Z_c = \frac{1}{s \cdot C} = \frac{314465.48}{s}$

$Z_L = 31{,}83\, s$

$$V_{out} = V_{in} \cdot \frac{Z_c + Z_L}{R + Z_c + Z_L}$$

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{Z_c + Z_L}{R + Z_c + Z_L} = \frac{\frac{1}{sC} + s.L}{R + \frac{1}{s \cdot C} + s \cdot L}$$

$$\frac{\frac{1}{sC} + s.L}{R + \frac{1}{sC} + s \cdot L} \cdot \frac{sc}{s \cdot c} \Rightarrow \frac{1 + s^2 LC}{s^2 LC + sRC + 1}$$

Para pasar a forma canonica

$$\frac{1 + s^2 LC}{s^2 LC + sRC + 1} \cdot \frac{\frac{1}{LC}}{\frac{1}{LC}} \Rightarrow \frac{s^2 + \frac{1}{LC}}{s^2 + s \cdot \frac{R}{L} + \frac{1}{LC}}$$ que es $$\frac{s^2 + w_0^2}{s^2 + \frac{w_0}{Q} + w_0^2}$$

$L \cdot C = 31{,}83 \cdot 10^{-3}\, H \cdot 3{,}18 \cdot 10^{-6}\, F = 1{,}0122 \cdot 10^{-7}$

$$\boxed{H(s) = \frac{s^2 + \frac{1}{1{,}0122 \cdot 10^{-7}}}{s^2 + s \cdot \frac{10^5}{31{,}83 \cdot 10^{-3}} + \frac{1}{1{,}0122 \cdot 10^{-7}}}}$$

Reemplazamos por "jw"

$$\boxed{H(jw) = \frac{-w^2 + \frac{1}{1{,}0122 \cdot 10^{-7}}}{-w^2 + jw \cdot \frac{10^5}{31{,}83 \cdot 10^{-3}} + \frac{1}{1{,}0122 \cdot 10^{-7}}}}$$

separamos parte real e imaginaria

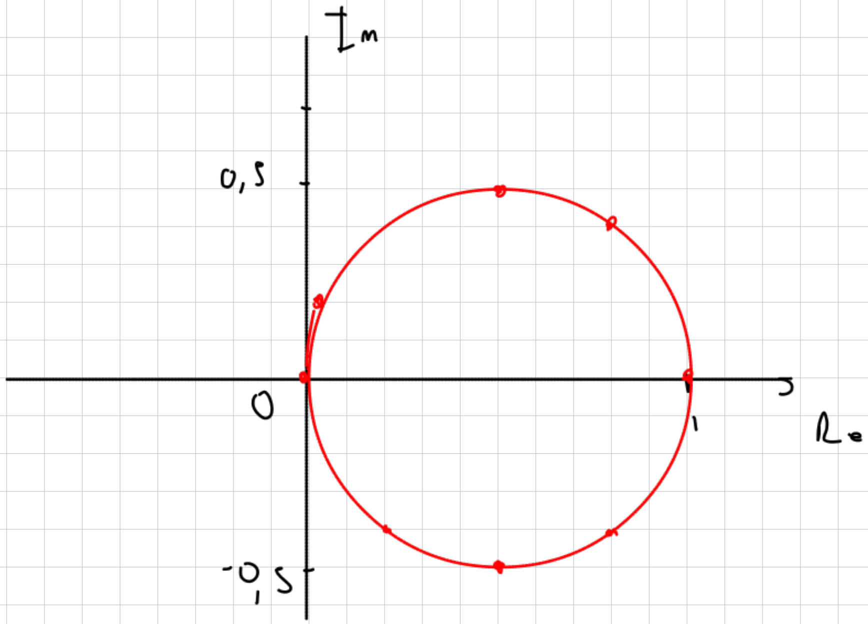
$$\frac{-w^2 + \frac{1}{1{,}0122 \cdot 10^{-7}}}{-w^2 + jw \cdot \frac{10^5}{31{,}83 \cdot 10^{-3}} + \frac{1}{1{,}0122 \cdot 10^{-7}}} \Rightarrow \frac{\overbrace{w_0^2 - w^2}^{A}}{w_0^2 - w^2 + j\underbrace{w\frac{R}{L}}_{B}} \Rightarrow \frac{A}{A + jB}$$

$$\frac{A}{A + jB} \cdot \frac{A - jB}{A - jB} = \frac{A^2 - jAB}{A^2 \cancel{- jAB} + \cancel{jAB} + B^2} = \frac{A^2}{A^2 + B^2} - j\,\frac{A.B}{A^2 + B^2}$$

$$Re\ \frac{(w_0^2 - w^2)^2}{(w_0^2 - w^2)^2 + \left(w\frac{R}{L}\right)^2} \qquad Im - \frac{(w_0^2 - w^2) \cdot w\frac{R}{L}}{(w_0^2 - w^2)^2 + \left(w\frac{R}{L}\right)^2}$$

diagrama polar

w	Re	Im
0	1	0
2π	0,2	-0,4
π	0,5	-0,5
$100.10^3.2\pi$	0,03	0,2
$100.10^4.2\pi$	0,8	0,4
$500.10^3.2\pi$	0,5	0,5
500.2π	≈ 0	≈ 0
$0,25.2\pi$	0,8	-0,4



$$\boxed{H(s) = \frac{s^2 + \frac{1}{1,0122.10^{-7}}}{s^2 + s.\frac{10^5}{31,83.10^{-3}} + \frac{1}{1,0122.10^{-7}}}}$$

Para Bode

$* = 9,879 .10^6 = p_1.p_2$

$$num = \frac{1}{1,0122.10^{-7}}^{*}\left(\frac{s^2}{\frac{1}{1,0122.10^{-7}}} + 1\right)$$

$$den = (s + 3,144).(s + 3,141.10^6)$$

$$\underbrace{3,144}_{p_1} . \left(\frac{s}{3,144} + 1\right) . \underbrace{3,141}_{p_2}.10^6 \left(\frac{s}{3,141.10^6} + 1\right)$$

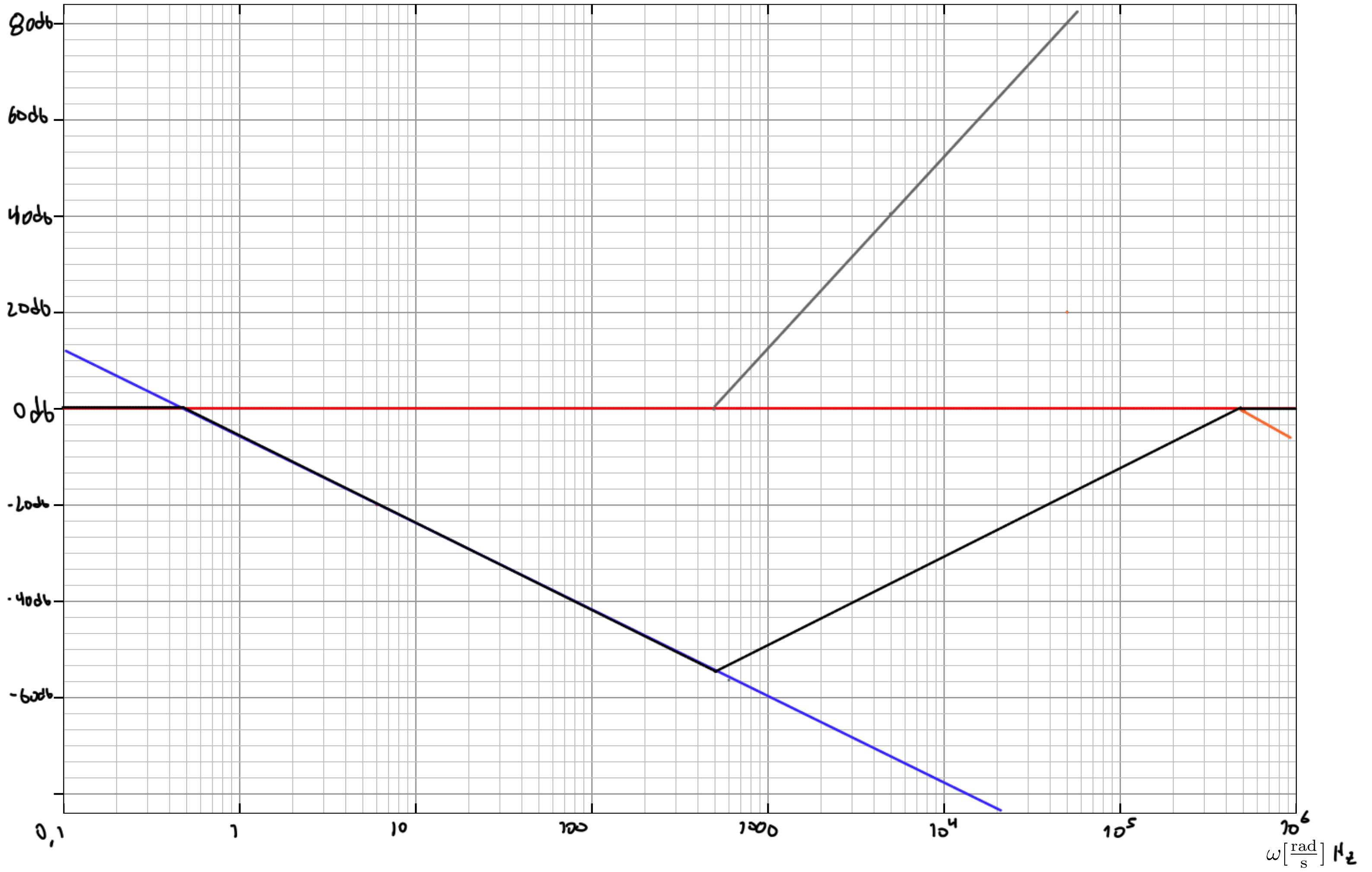
$$H(s) = \frac{\left(\frac{s^2}{9,874.10^6} + 1\right)}{\left(\frac{s}{3,144} + 1\right) . \left(\frac{s}{3,141.10^6} + 1\right)}$$

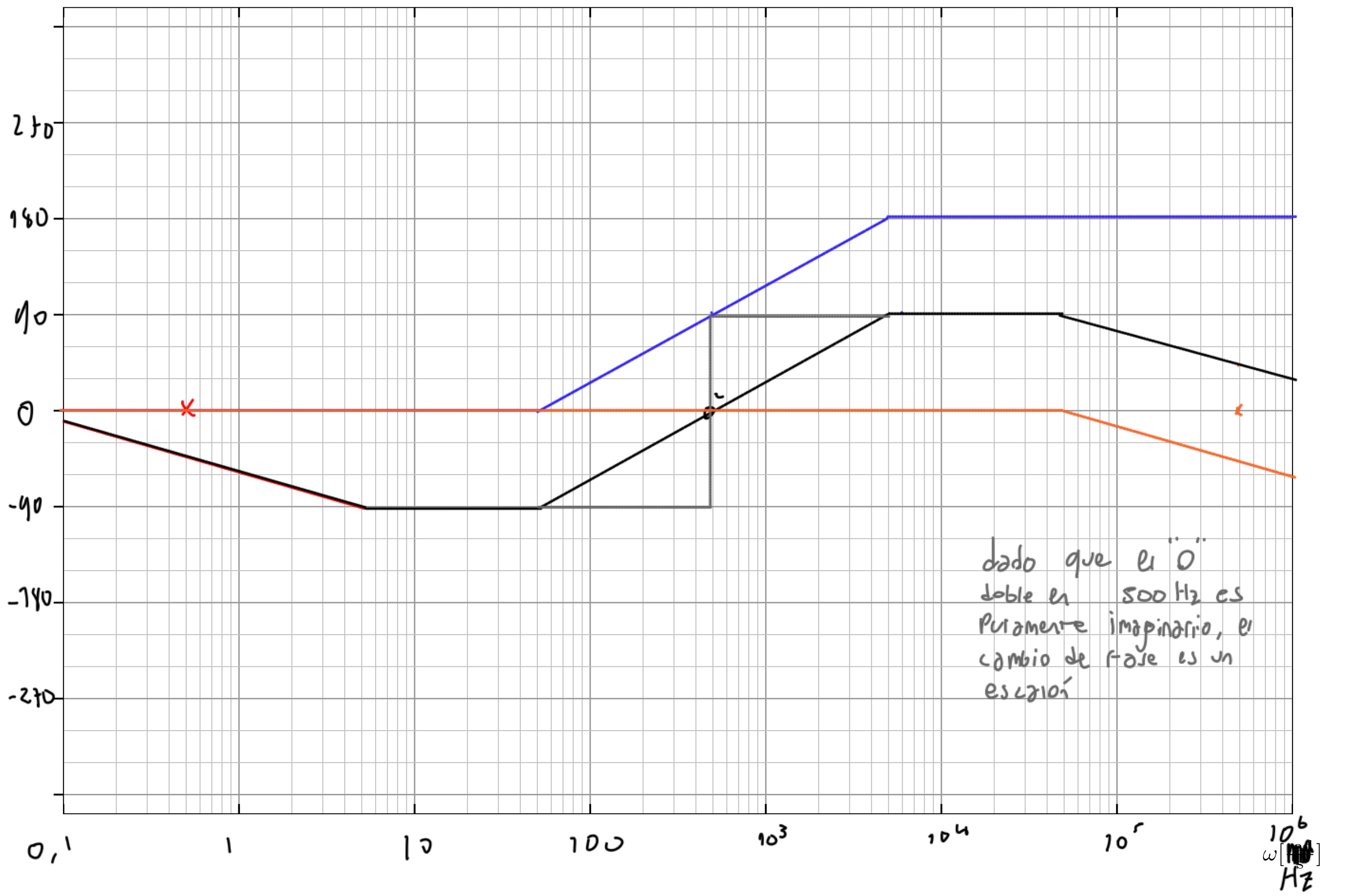
Cte = 1

Cero doble en $\frac{\sqrt{9,88.10^6}}{2\pi} = 500$ Hz

Polo simple en $\frac{3,144}{2\pi} = 0,5$ Hz

Polo simple en $\frac{3,141.10^6}{2\pi} \approx 500$ KHz


80db
60db
40db
20db
0db
-20db
-40db
-60db
0,1
1
10
100
1000
10^4
10^5
10^6
$\omega[\frac{\text{rad}}{\text{s}}]$ Hz


270
180
90
0
-90
-180
-270
0,1
1
10
100
10^3
10^4
10^5
10^6
Hz
dado que el "O" doble en 500 Hz es puramente imaginario, el cambio de fase es un escalón